

Ideological Expansion and Paradox

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1

Table of Contents

Parts

Mereology

Ideological Expansion and Paradox

Application: Proper Classes

Propositions

Quantified Propositional Logic

Ideological Expansion and Paradox

Application: Prior Kaplan

The Parallel

Conclusion

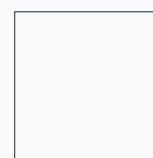
2

Parts

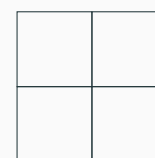
Parts

A relation of part to whole $\leq$ on U determines a Boolean algebra $(U, \leq)$ without a zero element:

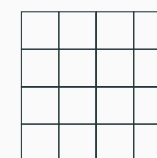
- There is a universal part U .
- If X is a part other than U , there is a complement $\bar{X}$.
- Two parts X and Y have a join $X \vee Y$ and a meet $X \wedge Y$.
- Join and meet distribute over each other.



$(U, \leq_0)$



$(U, \leq_1)$



$(U, \leq_2)$

3

What further parts we discern depends on what distinctions we are able to make. Given a condition $\varphi(u)$ in a given language $\mathcal{L}$:

- there is a join $\bigvee \{x \in U : \varphi(x)\}$.

The more distinctions we make, the more mereological structure we are able to discern.

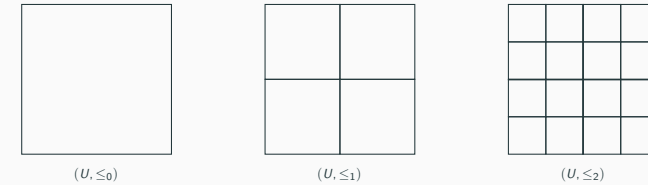
4

A Finest Decomposition?

$(U, \leq_\beta)$ refines $(U, \leq_\alpha)$ if, and only if,

- if $u \leq_\alpha v$, then $u \leq_\beta v$.
- some $\leq_\alpha$ -atoms are complex $\leq_\beta$ -fusions.

For example, each $(U, \leq_{n+1})$ is a refinement of its predecessor:



Is there a *finest* decomposition of the universe U into parts?

5

Tentative Suggestion

We will suggest that the answer to this question is negative.

If no partition is able to accommodate all the distinctions we might want to draw, then perhaps we consider an open-ended series of ever-finer partitions instead:

$$(U, \leq_0) \hookrightarrow (U, \leq_1) \hookrightarrow (U, \leq_2) \hookrightarrow \dots$$

Each partition refines the last, making distinctions to which earlier ones had been blind. But there is no finest partition of the domain.

We develop the paradox first, then return to this picture.

6

Classical Extensional Mereology (CEM)

CEM requires $\leq$ to be a partial order governed by the axioms:

$$\begin{aligned} x \not\leq y &\rightarrow \exists z (z \leq x \wedge \neg z \leq y) && \text{Strong Supplementation} \\ \exists x \varphi(x) &\rightarrow \exists x \text{Fu}(\langle \varphi \rangle, x) && \text{Fusion Schema} \end{aligned}$$

Notation:

$$\begin{aligned} x \circ y &:= \exists z (z \leq x \wedge z \leq y) \\ \text{Fu}(\langle \varphi(u) \rangle, x) &:= \forall y (\varphi(y) \rightarrow y \leq x) \wedge \\ &\quad \forall y (y \leq x \rightarrow \exists z (\varphi(z) \wedge u \circ z)) \end{aligned}$$

7

Models of CEM

Tarski (1956)

Complete Boolean algebras $(U, \sqsubseteq)$ without a zero element are models of CEM.

Completeness is overkill. Call a subset $X \subseteq U$ $\mathcal{L}^{\leq}$ -definable if X is the extension of some condition $\varphi(u)$ of $\mathcal{L}^{\leq}$ in the algebra.

Minimal Adequacy Constraint

A Boolean algebra $(U, \sqsubseteq)$ is a model of CEM if every $\mathcal{L}^{\leq}$ -definable subset $X \subseteq U$ has a join in the algebra.

8

Ideological Expansion

Expand the language with a new functional expression σ (for the singleton of, the essence of, etc.) governed by two axioms:

$$\begin{array}{ll} \exists x \sigma(x) \neq x & \text{Productivity} \\ \forall x \forall y (\sigma(x) \circ \sigma(y) \rightarrow x = y) & \text{Relative Atomicity} \end{array}$$

The singleton operation satisfies both constraints if, following [Lewis, 1991], singletons are mereological atoms.

9

Partial Consistency

CEM_0 is the restriction of CEM to the base language $\mathcal{L}^{\leq}$.

Partial Consistency

$\text{CEM}_0 + \text{Productivity} + \text{Relative Atomicity}$ is consistent.

Let $(U, \sqsubseteq)$ be a countable atomic Boolean algebra with a countable set of atoms A , which satisfies the minimal adequacy constraint. If $\pi : U \mapsto A$ is a bijection between U and A , then:

$(U, \sqsubseteq, \pi) \models \text{CEM}_0 + \text{Productivity} + \text{Relative Atomicity}$.

$(U, \sqsubseteq)$ is **not** complete; some sets of atoms **must** lack a join.

10

The Paradox

Inconsistency

$\text{CEM} + \text{Productivity} + \text{Relative Atomicity}$ is inconsistent.

Consider the condition of the expanded language $\mathcal{L}^{\leq, \sigma}$:

$$\exists x (u = \sigma(x) \wedge u \not\leq x)$$

- Some r fuses the condition. By Productivity, for some x , $x \neq \sigma(x)$. By Relative Atomicity, $\sigma(\sigma(x)) \not\leq \sigma(x)$.
- If r fuses the condition, then by logic alone, $\sigma(r) \leq r$.
- If $\sigma(r) \leq r$, then r overlaps some y such that $\sigma(y) \not\leq y$. Since $\sigma(y) \not\leq y$, $\sigma(y) \neq \sigma(r)$, and by Relative Atomicity, they do not overlap. Contradiction.

11

In what follows we interpret σ in terms of singleton.

We are not animated by a reductive aim in line with [Lewis, 1991], but rather by a potential response to the problem of proper classes.

For [Maddy, 1983], proper classes:

1. should be significantly different from sets, as opposed to another layer of the cumulative hierarchy.
2. should be “real, well-defined entities.”

[Gödel, 1964] articulated a contrast between combinatorial and logical collections. Here is the gloss [Maddy, 1983, 118] provides for the latter:

We imagine the entire universe separated into two heaps depending on whether or not things have the given property or fall under the given concept.

We identify classes with parts of the set-theoretic universe V .

A relation of part to whole on the set-theoretic universe $V(A)$ determines a Boolean algebra of classes $(\mathcal{C}, \sqsubseteq)$.

- There is a universal part of $V(A)$, the universal class.
- If X is a class other than $V(A)$, there is a complement $\bar{X}$.
- If X and Y are classes, there is a join $X \cup Y$ and a meet $X \cap Y$.
- Join and meet distribute over each other.

Given a condition $\varphi(u)$ in a given language $\mathcal{L}$,

- its instances have a join in the algebra.

Given a domain of classes $\mathcal{C}$ over $V(A)$, we find:

Remark

$(\mathcal{C}, \subseteq, \{\})$ **cannot** be a **complete** Boolean algebra if $\{\}$ verifies Productivity and Relative Atomicity.

Refinement

$$(\mathcal{C}, \sqsubseteq_0, \{\}) \hookrightarrow \dots (\mathcal{C}, \sqsubseteq_\alpha, \{\}) \hookrightarrow \dots$$

Let $\mathcal{L}_0$ be the language of set theory $\mathcal{L}_{ZF}$.

- $(\mathcal{C}, \sqsubseteq_0, \{\})$ incorporates a join for each condition $\varphi(x)$ formulated in $\mathcal{L}_0$. $\mathcal{C}$ consists of *predicative* classes over $V(A)$.

$\mathcal{L}_{\alpha+1}$ now expands $\mathcal{L}_\alpha$ with a predicate $\sqsubseteq^+$ for $\sqsubseteq_0$:

- $(\mathcal{C}, \sqsubseteq_{\alpha+1}, \{\})$ identifies a join for each *new* condition $\varphi(x)$ formulated in $\mathcal{L}_1$ with an erstwhile $\sqsubseteq_\alpha$ -atom.

Likewise for limit stages.

There is no finest partition of the domain into classes.

16

Propositions

Propositions

A partition of modal space determines a Boolean algebra of propositions $(B, \preceq)$ **with** a zero element:

- There is a universal proposition $\top$ and a trivial one $\perp$.
- The complement of a proposition p is the negation $\neg p$.
- Two propositions p and q have a join $p \vee q$ and a meet $p \wedge q$.
- Join and meet distribute over each other.

17

What further propositions we discern depends on what distinctions we are able to make. Given a condition $\varphi(p)$ in a given $\mathcal{L}$:

- there is a join $\bigvee \{p \in B : \varphi(p)\}$.

The more distinctions we make, the more propositional structure we should be able to discern.

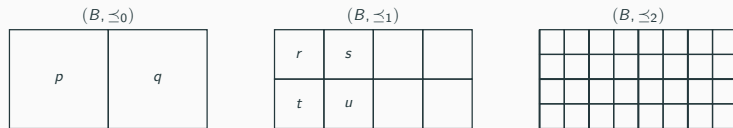
18

Is There a Finest Partition?

$(B, \preceq_\beta)$ refines $(B, \preceq_\alpha)$ if, and only if,

- if $p \preceq_\alpha q$, then $p \preceq_\beta q$.
- some $\preceq_\alpha$ -atoms are complex $\preceq_\beta$ -joins.

For example, each $(B, \preceq_{n+1})$ is a refinement of its predecessor:



Is there a *finest* partition of logical space into propositions?

19

Tentative Suggestion

We will suggest that the answer to this question is negative.

If no partition of modal space is able to accommodate all the distinctions we might want to draw, then perhaps we consider an open-ended series of ever-finer partitions instead:

$$(B, \preceq_0) \hookrightarrow \dots (B, \preceq_\alpha) \hookrightarrow \dots$$

Each partition refines the last, making distinctions to which earlier ones had been blind. But there is no finest partition of the domain.

We develop the paradox first, then return to this picture.

20

Quantified Propositional Logic

Let $\mathcal{L}_{QPL}$ be the language of quantified propositional logic:

	Expressions	Axioms
variables	$p, q, r, \dots$	
constants	$\perp, \top$	
connectives	$\neg, \wedge, \vee$	classical tautologies
quantifiers	$\forall, \exists$	elementary quantification theory
identity	$=$	reflexivity, Leibniz's Law

21

Boolean Structure

Given a Boolean algebra $\mathcal{B}$:

$$(B, \cap_{\mathcal{B}}, \cup_{\mathcal{B}}, \sim_{\mathcal{B}}, \top_{\mathcal{B}}, \perp_{\mathcal{B}}).$$

$$x \preceq_{\mathcal{B}} y := x \cap_{\mathcal{B}} y = x$$

An assignment α maps propositional variables into B :

$\llbracket p \rrbracket_\alpha$	$:= \alpha(p)$	
$\llbracket \top \rrbracket_\alpha$	$:= \top_{\mathcal{B}}$	top
$\llbracket \perp \rrbracket_\alpha$	$:= \perp_{\mathcal{B}}$	bottom
$\llbracket p \vee q \rrbracket_\alpha$	$:= \llbracket p \rrbracket_\alpha \cup_{\mathcal{B}} \llbracket q \rrbracket_\alpha$	join
$\llbracket p \wedge q \rrbracket_\alpha$	$:= \llbracket p \rrbracket_\alpha \cap_{\mathcal{B}} \llbracket q \rrbracket_\alpha$	meet
$\llbracket \neg p \rrbracket_\alpha$	$:= \sim_{\mathcal{B}} \llbracket p \rrbracket_\alpha$	complement

22

If $\mathcal{B}$ is a **complete Boolean algebra**, then:

$$\begin{aligned} \llbracket \exists p \varphi(p) \rrbracket_\alpha &:= \bigcup_{\mathcal{B}} \{ \llbracket \varphi \rrbracket_{\alpha[b/p]} : b \in B \} && \text{join} \\ \llbracket \forall p \varphi(p) \rrbracket_\alpha &:= \bigcap_{\mathcal{B}} \{ \llbracket \varphi \rrbracket_{\alpha[b/p]} : b \in B \} && \text{meet} \end{aligned}$$

Existential quantification is interpreted in terms of the join.

Universal quantification is interpreted in terms of the meet.

Consistency

A complete Boolean algebra $\mathcal{B}$ is a model for the axioms of QPL.

23

Expand the language with a propositional operator Q (for a propositional attitude) governed by two axioms:

$$\begin{aligned} \forall p (Qp \neq \perp) &&& \text{Regularity} \\ \forall p \forall q (Qp \wedge Qq \rightarrow p = q) &&& \text{Relative Atomicity} \end{aligned}$$

Interpret $Q\varphi$ as: φ is *uniquely queried* in line with [Kaplan, 1995].

24

The Paradox

Incompleteness

$(B, \cap_{\mathcal{B}}, \cup_{\mathcal{B}}, \sim_{\mathcal{B}}, \top_{\mathcal{B}}, \perp_{\mathcal{B}}, f)$ cannot be a **complete** Boolean algebra if f verifies Regularity and Relative Atomicity.

Define $R \subseteq B$ to be: $\{f(x) \cap \sim x : x \in B\}$.

- 1 If $\bigcup R$ exists, $\bigcup R \neq \perp$ By regularity, $f(\perp) \neq \perp$.
So, $\perp \prec f(\perp) \cap \sim \perp \preceq \bigcup R$
- 2 $f(\bigcup R) \cap \sim \bigcup R \preceq \bigcup R$ $f(\bigcup R) \cap \sim \bigcup R$ is in R
- 3 $f(\bigcup R) \cap \sim \bigcup R \preceq \sim \bigcup R$
- 4 $f(\bigcup R) \cap \sim \bigcup R = \perp$ from 2 and 3
- 5 $f(\bigcup R) \preceq \bigcup R$ from 4

25

- 6 $f(\bigcup R) \cap (f(x) \cap \sim x) \neq \perp$ Otherwise, $f(\bigcup R) \preceq \sim \bigcup R$.
for some $x \in B$. By 5, $f(\bigcup R) \preceq \bigcup R$. So
 $f(\bigcup R) = \perp$ pace Regularity.
- 7 $f(\bigcup R) \cap f(x) \neq \perp$ from 6
- 8 $\bigcup R = x$ Relative Atomicity, 7
- 9 $f(\bigcup R) \cap \sim x \neq \perp$ from 6
- 10 $\bigcup R \neq x$ from 4 and 9

We conclude that $\mathcal{B}$ is not a **complete** Boolean algebra.

26

Prior's Theorem

Prior's Theorem

$$\vdash Q\lambda \rightarrow \exists p(Qp \wedge p) \wedge \exists p(Qp \wedge \neg p)$$

where λ abbreviates $\exists p(Qp \wedge \neg p)$.

- | | | |
|---|--|--------------|
| 1 | $(Q\lambda \wedge \neg\lambda) \rightarrow \lambda$ | EG |
| 2 | $Q\lambda \rightarrow \lambda$ | from 1 |
| 3 | $Q\lambda \rightarrow (Q\lambda \wedge \lambda)$ | from 2 |
| 4 | $(Q\lambda \wedge \lambda) \rightarrow \exists p(Qp \wedge p)$ | EG |
| 5 | $Q\lambda \rightarrow \exists p(Qp \wedge p) \wedge \exists p(Qp \wedge \neg p)$ | from 3 and 4 |

27

From Prior to Kaplan

Corollary

QPL + Regularity + Relative Atomicity is inconsistent.

- | | | |
|---|-----------------------|-----------------------------------|
| 1 | $Q\lambda \neq \perp$ | Regularity |
| 2 | $Q\mu \wedge \mu$ | Prior's Theorem, 1, witness |
| 3 | $\lambda = \mu$ | Relative Atomicity, 1, 2 |
| 4 | λ | from 2, 3 |
| 5 | $Q\nu \wedge \neg\nu$ | witness from 4, def. of λ |
| 6 | $\lambda = \nu$ | Relative Atomicity, 1, 5 |
| 7 | $\perp$ | from 4, 5, 6 |

28

The Parallel

	Mereological	Propositional
Structure	Boolean algebra w/o $\perp$	Boolean algebra
Domain	parts	propositions
Operation f	$f(x) = \sigma(x)$	$f(p) = Qp$
Regularity	$\sigma(x)$ is non-null	$Qp \neq \perp$
Relative Atomicity	$\sigma(x) \circ \sigma(y) \rightarrow x = y$	$Qp \wedge Qq \rightarrow p = q$
Russell join	$\bigvee \{x : \sigma(x) \not\leq x\}$	$\exists p(Qp \wedge \neg p)$
Inconsistency	CEM + P + RA	QPL + R + RA

29

Whence the Contradiction?

CEM + Productivity + Relative Atomicity associates an object $\overline{\varphi(u)}$ with each formula $\varphi(u)$ of $\mathcal{L}^{\leq, \sigma}$, namely, the fusion of singletons of instances of $\varphi(u)$. This induces a satisfaction predicate:

$$\Sigma(u, \overline{\varphi(v)}) := \sigma(u) \leq \overline{\varphi(v)}.$$

Tarski's Undefinability Theorem

No formula $\Sigma(x, y)$ of $\mathcal{L}^{\leq, \sigma}$ is a satisfaction predicate for $\mathcal{L}^{\leq, \sigma}$.

For QPL + Regularity + Relative Atomicity, let $\overline{\varphi(q)}$ be $\exists p(Qp \wedge \varphi(p))$ and define:

$$\Sigma(q, \overline{\varphi(q)}) := Qp \rightarrow \overline{\varphi(q)}.$$

30

How to Respond

There are two main lines of response:

1. follow the arguments where they lead:

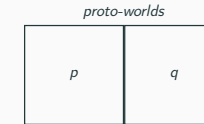
- no propositional operator verifies Regularity and Relative Atomicity on pain of contradiction; one **must** fail for *being uniquely queried*, e.g., [Bacon et al., 2016].

2. attempt to block the arguments:

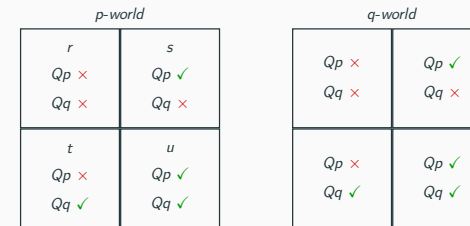
- impose some restriction on classical logic or existential generalization, e.g., [Kaplan, 1995].

31

Ontological Expansion: Ramification



p and q have level 0.



r, s, t, u have level 1.

Simplification: there should strictly be $2^4 = 16$ subdivisions to account for $Q\top$ and $Q\perp$ independently.

Subsequent partitions of modal space generate further higher-level propositions, which fall under an expanded domain of quantification.

32

Ramification

Propositions are generated in stages:

- Propositions of level 0 concern *extensional matters*, e.g., distributions of earth, air, fire, and water.
- Propositions of level 1 concern both extensional matters and patterns of application of Q to propositions of level 0.
- Propositions of level 2 include propositions of level 1 and patterns of application of Q to level 1.

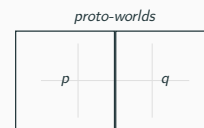
Once we ramify we write:

$$\exists p^n (Qp^n \wedge \neg p^n) \quad (\lambda^{n+1})$$

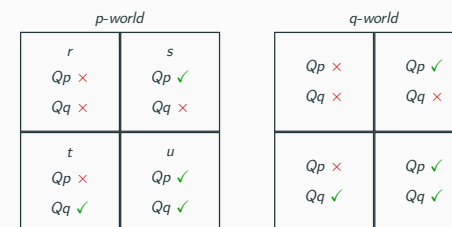
If λ is a proposition of level 1, then it states that at least one proposition of level 0 is both Q and false.

33

Ideological Expansion: Refinement



r, s, t, u remain indiscernible from p at stage 0.



r, s, t, u are discernible from p and one another at stage 1.

Simplification: there should strictly be $2^4 = 16$ subdivisions to account for $Q\top$ and $Q\perp$.

Subsequent refinements make further distinctions between propositions drawn from a *constant* domain.

34

Refinement

At stage α , we introduce a *pre-order* $\preceq_\alpha$ on a constant domain B , read: x entails y at stage α .

- $x \preceq_\alpha x$
- $x \preceq_\alpha y$ and $y \preceq_\alpha z$ implies $x \preceq_\alpha z$

$\preceq_0$ makes extensional distinctions between $\perp$, p , q , and $\top$. $\preceq_0$ remains blind to intensional distinctions drawn in terms of Q :

$$\begin{array}{ll} p \preceq_0 r & \text{and } r \preceq_0 p \\ r \preceq_0 t & \text{and } t \preceq_0 r \end{array}$$

At stage 1, we refine entailment to capture some such distinctions:

$$\begin{array}{ll} p \preceq_1 r & \text{but } r \not\preceq_1 p \\ r \not\preceq_1 t & \text{and } t \not\preceq_1 r \end{array}$$

35

α -Indiscernibility

Define:

$$x \equiv_\alpha y := x \preceq_\alpha y \wedge y \preceq_\alpha x$$

For example:

- p is, relative to $\preceq_0$, indiscernible from r , s , t , or u .
- If w will, at a later refinement, overlap₁ with both p and q , e.g., Qp , then w is, relative to $\preceq_0$, indiscernible from $\top$.

36

Quotient

The quotient $B/\equiv_\alpha$ inherits a Boolean structure from $\preceq_\alpha$:

$$\begin{array}{ll} [x]_\alpha \cup_\alpha [y]_\alpha & \text{join} \\ [x]_\alpha \cap_\alpha [y]_\alpha & \text{meet} \\ \sim_\alpha [x]_\alpha & \text{complement} \\ \top_\alpha & \text{top} \\ \perp_\alpha & \text{bottom} \end{array}$$

These operations take equivalence classes as inputs and outputs:

$$[p]_0 \cup_0 [q]_0 = [\top]_0.$$

37

Pull Back

Given a map $\pi_\alpha : B/\equiv_\alpha \rightarrow B$, we should be able to pull back to operations on B :

$$\begin{array}{ll} x \cup_\alpha y & := \pi_\alpha([x]_\alpha \cup_\alpha [y]_\alpha) \\ x \cap_\alpha y & := \pi_\alpha([x]_\alpha \cap_\alpha [y]_\alpha) \end{array}$$

One candidate for $\pi_\alpha([x])$ is the *weakest* proposition in $[x]$ by which we mean the proposition which will remain continue to entail the others through successive refinements of entailment, e.g., $\pi_\alpha([r])$ would be p .

38

$\exists p(Qp \wedge \neg p)$ will express different propositions relative to successive refinements of entailment.

Since $\preceq_0$ is blind to Q -distinctions, $Qx \equiv_0 \top$ for all $x \in B$.

$$\begin{aligned}\exists p(Qp \wedge_0 \neg_0 p) &= \bigcup_0 \{Qx \wedge_0 \sim_0 x : x \in B\} \\ &= \bigcup_0 \{\top \wedge_0 \sim_0 x : x \in B\} \\ &= \bigcup_0 \{\sim_0 x : x \in B\} \\ &= \top\end{aligned}$$

That means that $\lambda^0 = \top$.

One payoff of the refinement strategy is that it requires no revision of elementary quantification theory, since we operate within a constant domain of propositions.

The moral of the paradox is not that we omitted some propositions from the domain, e.g., existential generalizations expressed by a formula of the form $\exists p(Qp \wedge \neg p)$, but rather that ideological expansion increases our ability to make distinctions between erstwhile indiscernible propositions.

On the other hand, $\lambda^1 \prec_1 \top$.

$\preceq_1$ is sensitive to Q -distinctions for $\{\perp, p, q, \top\}$, yet $Qx \equiv_1 \top$ for all $x \in B \setminus \{\perp, p, q, \top\}$. Distinguish two cases:

Case 1: $x \in \{\perp, p, q, \top\}$.

$$\begin{aligned}Q\top \wedge_1 \sim_1 \top &= \perp & Qp \wedge_1 \sim_1 p &= Qp \wedge_1 q \\ Qq \wedge_1 \sim_1 q &= Qq \wedge_1 p & Q\perp \wedge_1 \sim_1 \perp &= Q\perp\end{aligned}$$

Case 2: $x \notin \{p, q, \top, \perp\}$.

$$Qx \wedge_1 \sim_1 x = \top \wedge_1 \sim_1 x = \sim_1 x$$

$$\lambda^1 = (Qp \wedge_1 q) \vee_1 (Qq \wedge_1 p) \vee_1 Q\perp \vee_1 \bigvee_1 \{\sim_1 x : x \notin \{p, q, \top, \perp\}\}$$

Conclusion

Ideological Expansion and Paradox

Complete Boolean algebras have been used to interpret:

- the language of mereology $\mathcal{L}_{CM}$, and
- the language of quantified propositional logic $\mathcal{L}_{QPL}$.

Ideological expansion threatens the prospects of such interpretations. No **complete** Boolean algebra interprets:

- an expansion of $\mathcal{L}_{CM}$ with a functional expression σ governed by Productivity and Relative Atomicity.
- an expansion of $\mathcal{L}_{QPL}$ with a propositional operator Q governed by Regularity and Relative Atomicity.

42

One common reaction is to fall back to ontological expansion.

- expand the domain of $\mathcal{L}_{CM}$ with new fusions for the problematic conditions, which is then closed under σ . $\leq$ is appropriately reinterpreted.
- ramify the domain of $\mathcal{L}_{QPL}$ with higher-level propositions for the problematic conditions, which is then closed under Q . $\leq$ is appropriately reinterpreted.

We have explored an alternative reaction:

- refine the interpretation of $\leq$ and identify the fusions of the problematic conditions with erstwhile atoms.
- refine the interpretation of entailment to make discriminations between propositions that had remained indiscernible relative to earlier interpretations of entailment.

43

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44

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45